Inhibition of Extracellular Enzyme Activity by Reactive Oxygen Species upon Oxygenation of Reduced Iron-bearing Minerals

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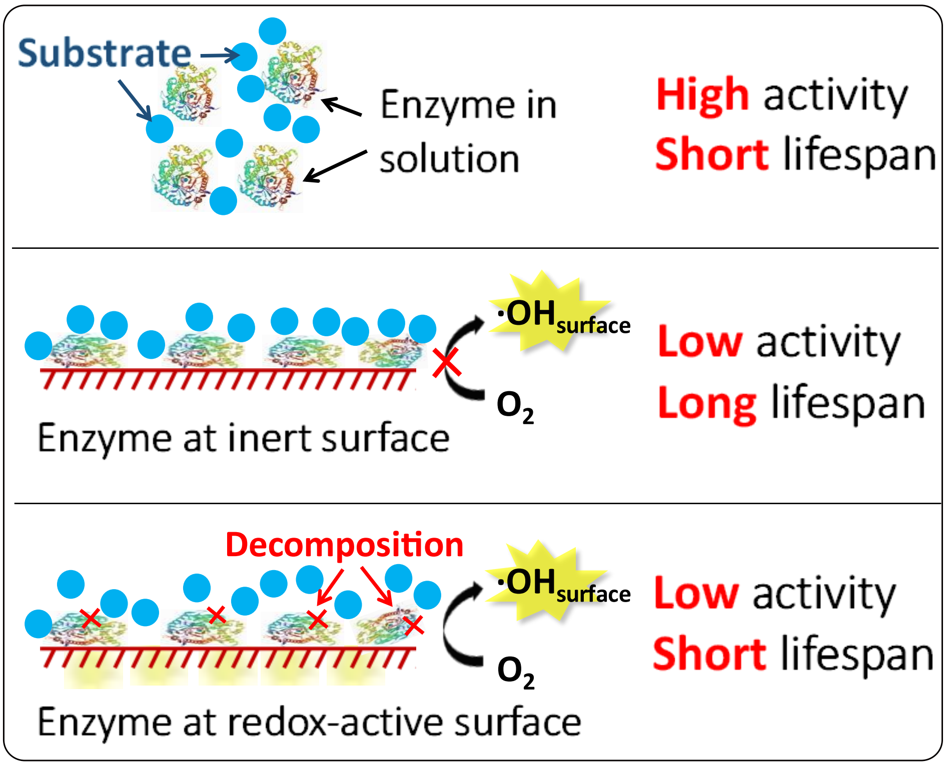

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ABSTRACT

The dual roles of minerals in inhibiting and prolonging extracellular enzyme activity in soils and sediments are governed by enzyme adsorption to mineral surface. Oxigenation of mineral-bound Fe(II) generates reactive oxygen species (ROS), yet it is unknown whether and how this process alters the activity and functional lifespan of extracellular enzymes. Here, the effect of mineral-bound Fe(II) oxidation on the hydrolytic activity of a cellulose-degrading enzyme β-glucosidase (BG) was studied using two pre-reduced Fe-bearing clay minerals (nontronite and montmorillonite) and one pre-reduced iron oxide (magnetite) at pH 5 and 7. Under anoxic condition, BG adsorption to mineral surface decreased its activity but prolonged its lifespan. Under oxic condition, ROS was produced, with the amount of •OH, the most abundant ROS, being positively correlated with the extent of structural Fe(II) oxidation in reduced minerals. •OH decreased BG activity and shortened its lifespan via conformational change and structural decomposition of BG. These results suggest that under oxic condition the ROS-induced inhibitory role of Fe(II)-bearing minerals outweighed their adsorption-induced protective role in controlling enzyme activity. These results disclose a previously unknown mechanism of extracellular enzyme inactivation, which have pivotal implications for predicting the active enzyme pool in redox-oscillating environments.

Keywords: β-glucosidase activity; clay mineral; Fe(II) oxidation; magnetite; reactive oxygen species;

Synopsis: ROS generated from oxygenation of structural Fe(II) in reduced minerals decreased β-glucosidase activity and shortened its lifespan.

INTRODUCTION

Extracellular enzymes, which catalyze the rate-limiting steps of carbon and nutrient cycling, exhibit variable activity and stability across soil and sedimentary ecosystems [1-5]. Free enzymes are commonly short-lived and sensitive to environmental perturbations [6, 7], including temperature [8], pH [9], soil organic matter (SOM) content [10], moisture [11], substrate availability [12], and other stresses [13, 14], whereas enzymes adsorbed onto mineral surface display repressed activity but longer functional lifespan. Mineral surface was historically considered as unreactive sorbent of enzymes [15, 16], such that the dual roles of minerals in inhibiting and prolonging enzyme activity were thought to depend on the binding capacity of enzymes to mineral surface [17]. Recent studies, however, have shown that mineral surface with catalytic properties (e.g., manganese oxides) can be effective agent of proteolysis [18, 19], and thus decrease the functional lifespan of enzymes [17]. Mineral surface can also catalyze chemical reactions, and thus plays an important role in transformation of SOM [20, 21]. However, the role of redox-active properties of minerals in regulating enzyme activity is poorly understood, thus hindering efforts of incorporating extracellular enzymes into biogeochemical models.

Fe oxides and Fe-bearing phyllosilicates (interchangeably called clay minerals) are common hosts of Fe in soil and sedimentary environments [22, 23], and can undergo redox cycles under O_2-fluctuating condition [24-26]. Under such conditions, the Fe and C redox cycles are often coupled across diverse environments including wetlands [27], rice paddy soils [28], and groundwater table fluctuating zones [29]. Upon oxygenation, structural Fe(II) in minerals is an efficient generator of reactive oxygen species (ROS) (such as $\cdot OH$, $\cdot O^{2-}$, and H_2O_2) and/or high valent Fe [i.e., Fe(IV)], which can decompose SOM and affect nutrient cycling [21, 30-33]. The generated reactive oxidants are considered harmful to organisms and thus inhibit cell growth [34-36]. As a defense mechanism, microorganisms can eliminate these oxidants via intracellular metalloenzymes [37].

There are both positive and negative effects of ROS on enzyme activity. ROS production from oxidation of dissolved Fe(II) has been shown to promote the activity of phenol oxidase, an oxidative enzyme [27, 38, 39], via stimulating its substrate (i.e., phenolics) decomposition or co-precipitation (with Fe oxides) [40]. In contrast, photochemically produced ROS from chromophoric dissolved organic matter can oxidize amino acids within enzyme structure [41], thus decreasing the activity of hydrolytic enzymes such as leucine aminopeptidase [42]. However, it is currently unknown whether reactive oxidants generated from mineral surface affect the activity of extracellular enzymes. Generation of oxidants from mineral-bound Fe(II) is more environmentally-relevant since this process is active over a large pH range across diverse soil and sedimentary ecosystems (pH 3-9) [30, 31].

For this reason, a mechanistic understanding of enzyme activity and functional lifespan upon oxidation of mineral-bound Fe(II) is highly desirable. We hypothesize that reactive oxidants decrease the activity and shorten the functional lifespan of extracellular enzymes; and the extent of these impacts depends on the amount of oxidants generated from mineral surface. To test this hypothesis, we chose β-glucosidase (BG) as a model enzyme, a well-known soil enzyme that catalyzes the final step in the breakdown of cellulose to glucose [9, 43-46]. It is soluble, exhibits β-strand and α-helix secondary structures, and unfolds upon adsorption to minerals. Fe-bearing minerals (clay minerals and iron-oxide) were pre-reduced in preparation for subsequent oxidation. The kinetics of Fe(II) oxidation and ROS generation, enzyme activity, and potential conformational changes of BG were measured. The results are profound in understanding enzyme-mineral interactions, with important implications for biogeochemical cycles of carbon and other elements in soil and sedimentary environments.

MATERIALS AND METHODS

Preparations of Enzyme, Chemicals, and Minerals

BG was used without any pretreatment (lyophilized powder, ≥2 units/mg, molecular weight=135 kDa, Sigma-Aldrich). 4-Methylumbelliferyl β-D-glucopyranoside (MUB-linked substrate) and 4-Methylumbelliferon (MUB) (standard for the catalytic product of MUB-linked substrate) were purchased from Sigma-Aldrich, USA. Clay minerals and Fe oxides are ubiquitous in soils and sediments [18]. Nontronite and montmorillonite are smectites with different Fe content and are capable of generating different amounts of oxidants upon oxygenation [31, 35]. Furthermore, these minerals do not adsorb sodium benzoate (SB), a ·OH quencher, which make it possible to quantify the role of ·OH in enzyme activity [21]. Fe-rich nontronite NAu-2 [21.2% total Fe, 98% of which is Fe(III)] and Fe-poor montmorillonite SWy-2 [2.3% total Fe, 97% of which is Fe(III)] were selected as model clay minerals. Magnetite (particle size >1 μm), which has >90% BG adsorption capacity [17], was chosen as a model iron oxide. The details for sample preparation and characterization were described in Supporting Information (SI) Section S1 and Table S1.

To prepare reduced minerals, structural Fe(III) was reduced to Fe(II) by the citrate-bicarbonate-dithionite (CBD) method (SI Section 1). After chemical reduction, mineral suspensions were centrifuged (13,000 g for 15 mins) and filtered through 0.22 μm nylon filters. Aqueous Fe(II) in the supernatant was negligible in reduced NAu-2 (hereafter as rNAu-2) and SWy-2 (rSWy-2), and <5% in reduced magnetite (rMagnetite) (SI Section S3). Aqueous Fe(II) was removed by multiple washes. X-ray powder diffraction (XRD) results (SI Section S2) showed that chemical reduction did not change the mineralogy (Figure S1).

After removal of aqueous Fe(II), the remaining solids were used for quantification of structural and adsorbed Fe(II) by sequential extraction (SI Section S3). The extraction results showed that 48.4% of Fe in rNAu-2 was Fe(II) [45.9% structural + 2.5% adsorbed], and 65.1% of Fe in rSWy-2 was Fe(II) [63.0% structural + 2.1% adsorbed]. While unreduced magnetite initially contained 26.8% Fe(II) (partially oxidized [47]), rMagnetite contained 31.1% structural + 0.8% adsorbed Fe(II). The reduced mineral solids were resuspended in acetate buffer inside an anaerobic chamber (98% N_2 and 2% H_2) to achieve pH 5 and 7 (Section S1). pH 5 was chosen because BG is most active at this pH [48], while pH 7 was used for a comparison but is also typical in many soils. Furthermore, Fe(II) oxidation and oxidant generation from reduced minerals are pH-dependent [31, 35], which is useful to exam the effect of oxidants on enzyme activity. In all subsequent experiments, pH was monitored and variation was within ± 0.15. All prepared mineral suspensions were stored in serum bottles under N_2 atmosphere.

Detection and Quantification of Oxidants from Fe(II) Oxidation

To initiate Fe(II) oxidation experiments, an aliquot of stock mineral suspensions was transferred to a fresh serum bottle in acetate buffer to yield 1 g/L suspension with air as the headspace gas. Throughout the Fe(II) oxidation period (24 h, the time when ROS generation mostly stops [34, 35]), change of Fe(II) speciation (including structural, adsorbed and soluble forms, SI Section 3) was monitored.

To quantify •OH production, aliquots of the rNAu-2, rSWy-2, and rMagnetite stocks were mixed with excess sodium benzoate (SB, 30 mM) and exposed to air for 24 hours (SI Section S4). •OH generation by total Fe(II) and structural Fe(II) was independently measured (Section SI S4). The oxidation products of SB, including 4-hydroxybenzoic acid (4-HBA), 3-hydroxybenzoic acid (3-HBA), salicylic acid (2-HBA), and dihydroxybenzoic acid (2,5-DHBA) were quantified with high-performance liquid chromatography (HPLC) to calculate •OH generation (SI Section S4).

H_2O_2 concentration was measured using the N,N-diethylphenylenediamine (DPD) method (SI Section S5). The generations of •OH and •O^{2-} during oxygenation of structural Fe(II) in rNAu-2 and rSWy-2 have been previously confirmed via electron paramagnetic resonance (EPR) in our previous studies [34, 35].

Mössbauer Spectroscopic Analysis

To further study Fe speciation during the Fe redox cycle of the minerals, Mössbauer spectroscopic analysis was conducted. In particular, Mössbauer spectroscopy is a promising tool for determination of Fe(IV) species [49, 50], another oxidant that may affect enzyme activity as well. Three forms of NAu-2 and magnetite, including unreduced, chemically (CBD)-reduced, and air re-oxidized samples, were analyzed at 30 K (for nontronite) and 125 K (for magnetite, slightly above Verwey transition temperature [51]) (SI Section S6).

Enzyme Adsorption and Activity Measurement

BG adsorption to mineral surface is the first step in its interaction with oxidants generated from oxygenation of mineral-associated Fe(II). Thus, the amount of BG adsorption onto both unreduced and reduced minerals was determined by mixing BG (ranging from 0.02 to 2 mg/L) and mineral suspension (1 g/L) under separate oxic and anoxic conditions. By using a constant mineral concentration but a range of BG concentration, the range of the BG/mineral weight ratio (0.02 to 2×10^{-3}) bracketed that of extractable protein concentration in natural soils (1 to 4×10^{-3} g per g soil) [52]. As a control, free BG activity (without minerals) was also measured. One hour was sufficient for BG adsorption to reach equilibrium [17, 53]. The amount of BG adsorption (mg) was quantified by subtracting the residual BG concentration in solution from the initial BG concentration (SI Section S7). To study the desorption of previously adsorbed BG, the supernatant in the equilibrated mineral-BG suspension was replaced with fresh acetate buffer [54, 55], and any desorbed BG concentration was measured.

The activity of BG was measured by the decomposition rate of its substrate via a fluorometric enzyme assay (excitation/emission 365/445 nm) under dark condition [56, 57]:

$$\text{Activity (nmol/hour)} = Buffer\ volume \times \frac{\frac{Assay - Mineral\ control}{Quench} - Substrate\ control}{Emission \times Time \times Assay\ volume}$$

Where, Buffer volume = volume of buffer used in enzyme-mineral suspension; Assay = average of “Mineral + Substrate” fluorescent value; Quench =

$\frac{Average\ mineral+Standard\ fluorescent\ values}{Average\ Standard\ fluorescent\ values}$; Substrate control = average of "Substrate" fluorescent value; Emission = $\frac{Average\ Standard\ fluorescent\ values}{nmol\ standard\ in\ assay}$; Time = incubation time in hours; Assay volume = volume of mineral suspension in each well. To measure BG activity, BG-mineral suspensions (total BG) and supernatant samples (supernatant BG) were seperately reacted with the MUB-linked substrate for 2 h in the black polystyrene 96-well plates (Corning®, USA) at room temperature. The catalytic product of the MUB-linked substrate, i.e., MUB, was measured using a microplate reader (Spectra Max ID3, USA). The sample layout in 96-well plates for different experimental settings was shown in SI Figure S2. BG activity was normalized relative to its mass (SI Section S7) [54, 55]. Because substrate adsorption to mineral surface can also affect enzyme activity [58], adsorptions of MUB-linked substrate and MUB standard to minerals were also examined (SI Section S8).

Impact of Fe(II) Oxidation on BG Activity

To initiate the Fe(II) oxidation experiment, BG was added to the pre-reduced mineral bottles, and the headspace of N_2 gas was replaced with air. Because the highest Fe(II) oxidation rate is typically observed in the first 2 to 4 h [21], BG was pre-exposed to air for 2 h, and BG activity was measured afterwards. Because Fe(II) oxidation and oxidant generation should have continued from 2 to 4 hours, the time period needed for enzyme activity measurement, both BG and substrate should have been exposed to the oxidants. In this case, altered BG activity may be due to the impacts of oxidants on both BG and the substrate. To eliminate this effect, BG activity was also measured after 24 h pre-exposure to oxidants, because at this point oxidant production should have stopped [21, 30, 31] and the substrate was no longer exposed to oxidants during BG activity measurement. Thus, 24-h measurement would have reflected the impact of oxidants on BG only, not on the substrate. Anaerobic experiments (i.e., N_2 as the headspace gas; degassed buffer as the working solution) were performed as controls. XRD was used to study whether there was any intercalation of BG or any decomposed products into the clay interlayer as a result of Fe(II) oxidation (SI Section S2). To separately assess if the substrate and standard were decomposed by ROS, the substrate and standard decomposition experiments were conducted (SI Section S9).

To further investigate the roles of different types of ROS in affecting enzyme activity, BG activity was measured after SB (30 mM), TEMPO (2 mM), and catalase (450 U/mL) were added to scavenge •OH, $•O^{2-}$, and H_2O_2, respectively [35]. Adsorption experiments confirmed that SB and TEMPO did not adsorb to the minerals under the experimental conditions (SI section S10). Furthermore, because of their excess amounts and interaction probability between mineral-adsorbed scavengers and mineral-produced ROS, adsorption of scavengers to mineral surface was not expected to significantly decrease their effectiveness. To eliminate any potential effects of scavengers on enzyme activity, control experiments were conducted in the presence of scavengers but without minerals.

Conformational Changes of β-glucosidase after Adsorption to Mineral Surfaces and after Exposure to ROS

Conformational changes of BG upon adsorption to mineral surfaces and after Fe(II) oxidation were examined using attenuated total reflection Fourier transform infrared spectroscopy (ATR-FTIR) with a Spectrum One Frontier FTIR spectrometer (Perkin-Elmer, Waltham, Massachusetts) (SI Section S11). Changes of BG adsorbed to mineral surface and after Fe(II) oxidation process were also probed by X-ray photoelectron spectroscopy (XPS) on a Kratos Ultra Axis Spectrometer (SI Section S12).

RESULTS AND DISCUSSION

Adsorption of β-glucosidase to Reduced Fe-bearing Minerals

The BG adsorption isotherms were dependent upon mineral types (Figure S3). BG adsorption to the two clays (>99% at both pH values) was higher than to the magnetite (>90% and >95% at pH 5 and 7, respectively), likely due to the much higher SSA of the clays (Table S1) [17, 59]. There was no intercalation of BG into the interlayer space of the clay minerals, as evidenced by lack of increase of d(001) spacing in XRD patterns (Figure S4), consistent with previous observations that proteins prefer to adhere to clay external surfaces when protein versus clay ratios are lower than 1: 4 [54, 60, 61].

Furthermore, the zeta-potentials of clay minerals were more negative than that of magnetite (Table S2). As a result, the calculated extent of opposite charge overlap between BG (positively charged due to the isoelectric point of 7.3 [44, 62]) and clay minerals (>90%) was higher than that between BG and magnetite (<76%) (Table S2). Nonetheless, surface-induced structural changes of BG may result in conformational entropy gains large enough to allow its adsorption to even “electrostatically hostile” surfaces [15, 16], resulting in >90% adsorption of BG to magnetite. Other non-electrostatic forces, such as ligand-metal binding between the carboxyl group of BG and hydroxyl group of Fe minerals may be important as well [18]. All these strong binding forces may have accounted for low levels of desorption in acetate buffer (<5%, Figure S5), in concert with other studies [54, 63, 64].

The extent of substrate/standard adsorption was negligible under the experimental condition (<4% in all cases, Figure S6), suggesting that the mobility of the substrate was not limited. Therefore, any change in BG activity should be due to the interaction between BG and mineral surface during adsorption and/or Fe(II) oxidation process, not due to any decreased probability of encounter between adsorbed BG and the substrate.

Generation of ROS upon Fe(II) Oxidation in Reduced Minerals

Under anoxic condition, there was no oxidation of structural Fe(II) in all reduced minerals at both pH values (Figure 1a, b). Under oxic condition, structural Fe(II) was oxidized, i.e., Fe(II) concentration significantly decreased under oxic condition than under anoxic condition (t-test, $p<0.001$, Table S3). The Fe(II)-rich rNAu-2 showed the highest extent of total Fe(II) oxidation, followed by the Fe(II)-poor rSWy-2 and rMagnetite (Figure 1a-b). Total and structural Fe(II) exhibited nearly identical oxidation patterns (Figures 1a-b & S7a-b), because adsorbed and soluble Fe(II) concentrations were negligible throughout the entire oxidation period (Figure S7c-f). Fe(II) oxidation was confirmed by Mössbauer spectroscopic analysis (Figure S8), showing an increased Fe(II)/total Fe ratios upon oxygenation of pre-reduced minerals.

The accumulative production pattern of •OH was similar to those of Fe(II) oxidation, exhibiting the order: rNAu-2 > rSWy-2 > rMagnetite (Figure 1 c, d). Similar production patterns were observed for H_2O_2 but with fluctuations (Figure S9). Significant positive correlations between the accumulative production of •OH and the extent of Fe(II) oxidation illustrated that Fe(II) oxidation was responsible for the production of •OH (Figure 1e, f). Although the extent of Fe(II) oxidation was higher at pH 7 than at pH 5 (Figure 1a, b), the •OH and H_2O_2 yields were actually higher at pH 5 (Figure 1c, d), because ROS generation was favored under acidic condition [31, 34].

Owing to the low levels of mineral-adsorbed and soluble Fe(II) (Figure S7c-f), ROS should be predominantly generated from oxygenation of structural Fe(II), as evidenced by the same yield of •OH from structural Fe(II) and total Fe(II) (t-test, $p>0.05$) (Figure S10). In addition to ROS, oxidation of Fe(II) may form Fe(IV), especially from soluble Fe(II) [35]. However, Mössbauer spectra of rNAu-2 and rMagnetite (Figure S8) did not detect any characteristic Fe(IV) signals [49, 50], consistent with the low level of soluble Fe(II) in all treatments (Figure S7e-f).

In terms of production efficiency of •OH [defined as •OH yield (μM) per mM Fe(II) oxidized], rSWy-2 was the highest (the highest slope in the correlation results in Figure 1e-f). At a given mineral concentration, structural Fe(II) in Fe(II)-poor rSWy-2 should be distributed more distantly from one another than that in Fe(II)-rich rNAu-2, hence resulting in more scattered ROS production sites and a less self-quenching effect [31]. Furthermore, the generation of •OH depends on the mineral crystal chemistry and density of •OH production sites such as the metal-oxygen bonds [31, 65], which may be different between rSWy-2 and the other two minerals.

Effect of Fe(II) Oxidation on β-glucosidase Activity

In the absence of minerals, the activity of free BG decreased by 50% from pH 5 to 7 (optimal pH for BG is 4-6 [48]) and by 30% from 2 to 24 hours (t-test, $p<0.05$, Table S4), but did not show difference under oxic and anoxic conditions (Figures 2).

In the presence of minerals, the enzyme activity has two dimensions of interest: a) absolute decrease in enzyme activity (hereafter simply termed activity for brevity), and b) time it takes for enzyme activity to equilibrate at a reduced level (functional lifespan) [17, 66]. Both dimensions were sensitive to redox condition and pH (Figures 2, S11 and S12 for rNAu-2, rSWy-2, and rMagnetite, respectively). Under anoxic condition, the decreased activity due to the presence of minerals (Figure 2) was ascribed to BG adsorption onto mineral surface [17]. However, the mineral-adsorbed BG activity did not decrease from 2 h to 24 h ($p>0.05$, Table S4), in contrast to free BG. This result suggested that BG adsorption to mineral surface under anoxic condition decreased its activity, but increased its functional lifespan. This protective role of reduced mineral surface under anoxic condition was similar to the role of unreduced and redox-inert minerals under oxic condition [17, 66].

Under oxic condition, ROS were produced from oxygenation of structural Fe(II) in the reduced minerals. Due to their non-selective nature, these surface ROS would attack the materials (e.g., enzyme and substrate) in the solid-liquid boundary layer and adsorbed on the mineral surface [67], thus affecting enzyme activity [21, 34, 35]. However, ROS did not decompose the substrate (Figure S13). Therefore, under oxic condition the substrate remained intact and BG was available to interact with it (i.e., BG adsorption on external surface, not intercalated into clay interlayers, Figure S4), any change in BG activity should be attributed to alteration of the BG structure, as described below.

The oxidation of Fe(II) in the reduced minerals decreased the activity of mineral-adsorbed BG (Figures 2, S11 & S12). This effect was evidenced by the significant difference in BG activity between anoxic and oxic conditions (solid vs. hollow symbols) for all three minerals ($p<0.05$, Table S4), while especially true for rNAu-2 (Figure 2). A longer exposure time to ROS decreased the BG activity to a greater extent, i.e., the activity was significantly lower in 24-hour relative to that in 2-hour (Table S4), suggesting that the lifespan of BG was shortened by ROS. These results challenge previous understandings that minerals normally prolong enzyme lifespan [5, 17, 66].

The extent of BG activity inhibition by ROS was mineral-dependent. rNAu-2 exhibited the highest level of inhibition (up to 5-fold from comparing the activity under anoxic and oxic conditions) (Figure 2b, d), in concert with its highest ROS generation (Figure 1). However, the amount of inhibition (i.e., $\text{Activity}_{\text{oxic}}/\text{Activity}_{\text{anoxic}}$) per unit of oxidized Fe(II) was greater by rSWy-2 than by rNAu-2 and rMagnetite (i.e., slopes in Figure 3a), consistent with the higher production efficiency of ·OH by Fe-poor rSWy-2 (i.e., ·OH yield per unit of Fe(II) oxidized, slopes in Figure 1e-f). The significant linear correlations between the inhibition extent of BG activity ($\text{Activity}_{\text{oxic}}/\text{Activity}_{\text{anoxic}}$) and the cumulative amount of •OH (Figure 3b) unraveled that •OH contributed to the observed BG activity suppression, likely owing to its strong oxidation ability (Eh = 2.8 V) [21, 68]. In addition to the mineral-adsorbed BG activity, un-adsorbed aqueous BG (<10%, Figure S3) in the rMagnetite treatment also declined after its exposure to ROS (Figure S14).

To further identify the role of individual ROS in enzyme activity, scavengers were added (SB for •OH, catalase for H_2O_2, and TEMPO for $•O^{2-}$). Scavengers themselves did not inhibit enzyme activity (Figure S15). Consistent patterns were observed at both pH values and two time points across all three minerals (Figure 3c-d for rNAu-2; Figure S16 for rSWy-2 and rMagnetite). Addition of SB almost completely removed the inhibition of BG activity by reduced minerals at both pH 5 and 7 (Figures 3c-d), confirming that •OH was predominantly responsible for BG activity suppression. This result was evidenced by the significant difference between the “SB” and no scavenger “rNAu-2” groups under oxic condition ($p<0.05$) but the insignificant difference between the “SB” and no scavenger “rNAu-2” groups under anoxic condition ($p>0.05$) (Table S5). The greatest inhibitory role of •OH was in agreement with the predominant antimicrobial role of •OH [34, 35]. Among the three minerals, SB was most effective in removing the inhibitory effect of rNAu-2, consistent with the highest amount of •OH production by this mineral. In comparison, catalase addition partially removed the inhibitory effect of rNAu-2 (Figure 3c), in line with the highest H_2O_2 production by rNAu-2 (Figure S9). TEMPO addition also removed some level of the inhibitory effect of rSWy-2 (Figure S16a), indicating the possible effect of $•O^{2-}$ on BG activity.

The effects of Fe(II)-induced ROS on enzyme activity have been documented in natural environments but only indirect pathways are described. For instance, the activity of BG is positively correlated with soluble or HCl-extractable Fe(II) concentration in some soils and sediments [27, 69]. This correlation is likely caused by a combined effect of ROS-induced degradation of phenolics [40] and its co-precipitation with short-range-ordered Fe oxides [70], thereby reducing the toxicity of phenolics to BG and increasing its activity [71]. Additionally, phenolics that have been regarded as toxic materials to BG do not always accumulate in soils [72], thus the factors manipulating BG activity may not be limited to this indirect pathway. In the current study, the predominant mechanism of BG inactivation was via attack by ROS produced from oxidation of structural Fe(II) in minerals, uncovering a direct pathway for enzyme inactivation. These Fe-bearing minerals, particularly clays, can be cycled multiple times without losing their capacity for ROS generation [30]. Thus, redox-active Fe-bearing minerals likely play a significant role in regulating enzyme activity and functional lifespan.

Conformational and Structural Changes of BG upon Fe(II) Oxidation

Unlike small and rigid molecules or ions, enzymes undergo irreversible conformational change upon adsorption onto mineral surface [17], which affects enzyme activity. ATR-FTIR spectra of BG adsorbed on minerals comprised the characteristic bands of Amide I, Amide II, Amide III, and side-chain δ CH_2 and υs COO- [73] (Figure 4a). Amongst these characteristic peaks, Amide I band (1700-1600 cm^{-1}), mainly the result of carbonyl stretching vibrations of the C=O bond of amide, is commonly considered the optimal predictor for quantitating the secondary structures of proteins [9, 55, 64, 73]. Hence, it was deconvoluted into α-helices, β-sheets, β-turns, and random coils [73] (Figure 4b). In comparison to free BG ("Control" in Figure 4c), the relative proportions of these four secondary structures of Amide I varied after BG adsorption to the unreduced minerals (Figure 4c). Such conformational change was previously documented to be a function of the surface area-normalized amount of adsorption [17]. In comparison to the free BG and unreduced minerals, the significantly decreased α-helix proportion for adsorbed BG to rNAu-2 and rMagnetite upon Fe(II) oxidation suggested BG unfolding, which may be a cause for declined activity [15, 17]. Considering that multiple mineral surface properties, including electric properties (Table S2) and surface area [35] all changed after reduction, this type of BG unfolding upon Fe(II) oxidation could be attributed to a combined effect of altered surface property during chemical reduction and ROS attack.

Relative to the reduced minerals without scavenger, addition of SB significantly increased the proportion of α-helix for all three minerals ($p<0.05$), hinting at a refolding of BG due to quenching of •OH by SB [55, 64]. The proportion of α-helix increased more after scavenging of •OH than after scavenging of H_2O_2 and O^{2-}, suggesting that •OH was likely the main ROS in causing unfolding of the BG molecule. The protein-mineral interaction would result in a loss in the helical structure (α-helix) and increase in random coils and β-sheets, while the intensity of such interaction force was tightly related to particle size, surface area, hydrophobicity, and polarity of a mineral surface [15, 17, 64]. Our study suggests that the redox activity of mineral surface can also induce potential enzyme conformational changes.

XPS may be able to probe the variations of aliphatic and carbonyl carbons within the mineral-adsorbed BG molecule [9]. Relative to BG adsorbed to the unreduced mineral surfaces, BG molecule adsorbed to rNAu-2 and rSWy-2 surfaces upon Fe(II) oxidation displayed lower proportions of C-N/C-O and C=O (Figure S17), and rMagnetite displayed decreased C-C/C-H and C=O proportions. These functional groups were likely derived from the characteristic amide groups of the BG molecule [9]. Relative to the reduced minerals, addition of SB (•OH scavenger) enhanced the C-N/C-O fraction for all three reduced minerals and C=O fraction for rMagnetite, again underpinning the role of •OH in the BG molecular structure. These results evinced the selective attack of ROS on the BG molecular structure, yet this attack might build upon the initial conformational change of BG adsorbed to mineral surfaces (involves a whole cascade of angular rotation or structural reorganization [16]). The effect of ROS generated via mineral-bound Fe(II) oxidation on BG decomposition is analogous to its role in promoting SOM decomposition [21, 28, 39, 74], ultimately resulting in enzyme inactivation.

ENVIRONMENTAL IMPLICATIONS

Minerals as adsorbents have been demonstrated to play dual roles in inhibiting and protecting extracellular enzyme activity: (a) decrease in enzyme activity and (b) prolonged functional lifespan [17, 66]. Because most Fe(II) in soils and sediments occurs as mineral-bound form [75, 76] and Fe-bearing minerals can undergo multiple Fe redox cycles in soil and sedimentary environments [22, 25, 77], our results disclose a previously unknown mechanism of enzyme inactivation. Extracellular enzymes first bind to mineral surface, initially resulting in a decreased activity. Subsequently, the roles of minerals in altering the enzyme functional lifespan depend on three circumstances: (1) If the mineral surface is redox inert, relative to free enzyme, the functional lifespan of mineral-adsorbed enzyme is prolonged; (2) If the mineral surface is redox-active, the enzyme functional lifespan is still prolonged under anoxic condition; (3) If Fe(II)-rich minerals are exposed to air under oscillating redox condition, ROS-induced BG decomposition further inactivates enzyme activity and shortens enzyme lifespan. Under the latter two circumstances, the interactions between Fe minerals and enzymes may be long-lasting over multiple Fe redox cycles.

Mineral-enzyme interactions typically occur at a microscopic scale but the effect is often profound at the global scale. The association of SOM with minerals has been proposed or proven to protect SOM from degradation via physical-chemical protection mechanisms [18, 78], however, there is the other side of the coin, i.e., a significant amount of SOM decomposition may occur in redox-active environments due to ROS attack upon oxygenation of structural Fe(II) in minerals [21, 33]. Given the relevance of extracellular enzymes to SOM decomposition and biogeochemical cycles of other elements, our results demonstrate that Fe-bearing minerals can significantly decrease enzyme activity, and thus offer additional protection to SOM. In this light, our study suggests that it is pivotal to consider the inhibitory effect of the redox-active minerals on SOM-degrading extracellular enzyme activities, which would, in turn, affect SOM turnover at a global scale. Therefore, the ultimate fate of SOM depends on a fine balance between protection (through physical/chemical protection of SOM and inhibition of extracellular enzyme activities) and destruction by ROS [21, 39].

ASSOCIATED CONTENT

Supporting Information

Detailed experimental methods (Sections S1−S12); Additional Tables S1−S7 and Figures S1-S17.

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REFERENCES

1. Allison, S. D., Cheaters, diffusion and nutrients constrain decomposition by microbial enzymes in spatially structured environments. *Ecology Letters* **2005,** *8*, (6), 626-635.
2. Burns, R. G.; DeForest, J. L.; Marxsen, J.; Sinsabaugh, R. L.; Stromberger, M. E.; Wallenstein, M. D.; Weintraub, M. N.; Zoppini, A., Soil enzymes in a changing environment: Current knowledge and future directions. *Soil Biology and Biochemstry* **2013,** *58*, 216-234.
3. Sinsabaugh, R. L.; Hill, B. H.; Shah, J. J. F., Ecoenzymatic stoichiometry of microbial organic nutrient acquisition in soil and sediment. *Nature* **2009,** *462*, (7274), 795-U117.
4. Sinsabaugh, R. L.; Lauber, C. L.; Weintraub, M. N.; Ahmed, B.; Allison, S. D.; Crenshaw, C.; Contosta, A. R.; Cusack, D.; Frey, S.; Gallo, M. E.; Gartner, T. B.; Hobbie, S. E.; Holland, K.; Keeler, B. L.; Powers, J. S.; Stursova, M.; Takacs-Vesbach, C.; Waldrop, M. P.; Wallenstein, M. D.; Zak, D. R.; Zeglin, L. H., Stoichiometry of soil enzyme activity at global scale. *Ecology Letters* **2008,** *11*, (11), 1252-1264.
5. Allison, S. D., Soil minerals and humic acids alter enzyme stability: implications for ecosystem processes. *Biogeochemistry* **2006,** *81*, (3), 361-373.
6. Luo, L.; Meng, H.; Gu, J. D., Microbial extracellular enzymes in biogeochemical cycling of ecosystems. *Journal of Environmental Management* **2017,** *197*, 539-549.
7. Sinsabaugh, R. L.; Follstad Shah, J. J., Ecoenzymatic stoichiometry and ecological theory. *Annual Review of Ecology, Evolution, Systematics* **2012,** *43*, 313-343.
8. Steinweg, J. M.; Dukes, J. S.; Paul, E. A.; Wallenstein, M. D., Microbial responses to multi-factor climate change: effects on soil enzymes. *Frontiers in microbiology* **2013,** *4*.
9. Serefoglou, E.; Litina, K.; Gournis, D.; Kalogeris, E.; Tzialla, A. A.; Pavlidis, I. V.; Stamatis, H.; Maccallini, E.; Lubomska, M.; Rudolf, P., Smectite clays as solid supports for immobilization of beta-

glucosidase: Synthesis, characterization, and biochemical properties. *Chemistry of Materials* **2008,** *20*, (12), 4106-4115.
10. Cenini, V. L.; Fornara, D. A.; McMullan, G.; Ternan, N.; Carolan, R.; Crawley, M. J.; Clement, J. C.; Lavorel, S., Linkages between extracellular enzyme activities and the carbon and nitrogen content of grassland soils. *Soil Biology abd Biochemistry* **2016,** *96*, 198-206.
11. Geisseler, D.; Horwath, W. R.; Scow, K. M., Soil moisture and plant residue addition interact in their effect on extracellular enzyme activity. *Pedobiologia* **2011,** *54*, (2), 71-78.
12. Zhou, L. H.; Liu, S. S.; Shen, H. H.; Zhao, M. Y.; Xu, L. C.; Xing, A. J.; Fang, J. Y., Soil extracellular enzyme activity and stoichiometry in China's forests. *Functional Ecology* **2020,** *34*, (7), 1461-1471.
13. Aponte, H.; Meli, P.; Butler, B.; Paolini, J.; Matus, F.; Merino, C.; Cornejo, P.; Kuzyakov, Y., Meta-analysis of heavy metal effects on soil enzyme activities. *Science of the Total Environment* **2020,** *737*.
14. Riah, W.; Laval, K.; Laroche-Ajzenberg, E.; Mougin, C.; Latour, X.; Trinsoutrot-Gattin, I., Effects of pesticides on soil enzymes: a review. *Environmental Chemistry Letters* **2014,** *12*, (2), 257-273.
15. Norde, W., My voyage of discovery to proteins in flatland... and beyond. *Colloids surfaces. B, Biointerfaces* **2008,** *61*, (1), 1-9.
16. Rabe, M.; Verdes, D.; Seeger, S., Understanding protein adsorption phenomena at solid surfaces. *Advances in colloid interface science* **2011,** *162*, (1-2), 87-106.
17. Sheng, Y.; Dong, H.; Coffin, E.; Myrold, D.; Kleber, M., The important role of enzyme adsorbing capacity of soil minerals in regulating β-glucosidase activity. *Geophysical Research Letters* **2022,** *49*, (6), e2021GL097556.
18. Kleber, M.; Bourg, I. C.; Coward, E. K.; Hansel, C. M.; Myneni, S. C. B.; Nunan, N., Dynamic interactions at the mineral–organic matter interface. *Nature Reviews Earth & Environment* **2021**.
19. Chacon, S. S.; Reardon, P. N.; Burgess, C. J.; Purvine, S.; Chu, R. K.; Clauss, T. R.; Walter, E.; Myrold, D. D.; Washton, N.; Kleber, M., Mineral surfaces as agents of environmental proteolysis: mechanisms and controls. *Environment Science & Technology* **2019,** *53*, (6), 3018-3026.
20. Shi, L.; Dong, H. L.; Reguera, G.; Beyenal, H.; Lu, A. H.; Liu, J.; Yu, H. Q.; Fredrickson, J. K., Extracellular electron transfer mechanisms between microorganisms and minerals. *Nature Reviews Microbiology* **2016,** *14*, (10), 651-662.
21. Zeng, Q.; Wang, X.; Liu, X.; Huang, L.; Hu, J.; Chu, R.; Tolic, N.; Dong, H., Mutual interactions between reduced Fe-bearing clay minerals and humic acids under dark, oxygenated conditions: Hydroxyl radical generation and humic acid transformation. *Environment Science & Technology* **2020,** *54*, (23), 15013–15023.
22. Dong, H. L.; Jaisi, D. P.; Kim, J.; Zhang, G. X., Microbe-clay mineral interactions. *American Mineralogist* **2009,** *94*, (11-12), 1505-1519.
23. Dong, H.; Huang, L.; Zhao, L.; Zeng, Q.; Liu, X.; Sheng, Y.; Shi, L.; Wu, G.; Jiang, H.; Li, F., A critical review of mineral–microbe interaction and co-evolution: mechanisms and applications. *National Science Review* **2022,** *9*, (10), nwac128.
24. Kappler, A.; Bryce, C.; M., M.; Lueders, U.; Byrne, J. M.; Swanner, E. D., An evolving view on biogeochemical cycling of iron. *Nature Reviews Microbiology* **2021,** *19*, 360–374.
25. Zhao, L.; Dong, H.; Kukkadapu, R. K.; Zeng, Q.; Edelmann, R. E.; Pentrak, M.; Agrawal, A., Biological redox cycling of iron in nontronite and its potential application in nitrate removal. *Environment Science & Technology* **2015,** *49*, (9), 5493-5501.
26. Sheng, Y.; Dong, H.; Kukkadapu, R. K.; Ni, S.; Zeng, Q.; Hu, J.; Coffin, E.; Zhao, S.; Sommer, A.

J.; McCarrick, R. M., Lignin-enhanced reduction of structural Fe (III) in nontronite: Dual roles of lignin as electron shuttle and donor. *Geochimica et Cosmochimica Acta* **2021**.

27. Wang, Y. Y.; Wang, H.; He, J. S.; Feng, X. J., Iron-mediated soil carbon response to water-table decline in an alpine wetland. *Nature Communications* **2017,** *8*.

28. Hanke, A.; Cerli, C.; Muhr, J.; Borken, W.; Kalbitz, K., Redox control on carbon mineralization and dissolved organic matter along a chronosequence of paddy soils. *European Journal of Soil Science* **2013,** *64*, (4), 476-487.

29. Briggs, M. A.; Day-Lewis, F. D.; Zarnetske, J. P.; Harvey, J. W., A physical explanation for the development of redox microzones in hyporheic flow. *Geophysical Research Letters* **2015,** *42*, (11), 4402-4410.

30. Zeng, Q.; Dong, H. L.; Wang, X.; Yu, T.; Cui, W. H., Degradation of 1, 4-dioxane by hydroxyl radicals produced from clay minerals. *Journal of Hazardous Materials* **2017,** *331*, 88-98.

31. Wang, X.; Dong, H. L.; Zeng, Q.; Xia, Q. Y.; Zhang, L. M.; Zhou, Z. Q., Reduced iron-containing clay minerals as antibacterial agents. *Environment Science & Technology* **2017,** *51*, (13), 7639-7647.

32. Yuan, S.; Liu, X.; Liao, W.; Zhang, P.; Wang, X.; Tong, M., Mechanisms of electron transfer from structrual Fe(II) in reduced nontronite to oxygen for production of hydroxyl radicals. *Geochimica et Cosmochimica Acta* **2018,** *223*, 422-436.

33. Yu, G.-H.; Kuzyakov, Y., Fenton chemistry and reactive oxygen species in soil: Abiotic mechanisms of biotic processes, controls and consequences for carbon and nutrient cycling. *Earth-Science Reviews* **2021,** *214*, 103525.

34. Xia, Q.; Wang, X.; Zeng, Q.; Guo, D.; Zhu, Z.; Chen, H.; Dong, H., Mechanisms of enhanced antibacterial activity by reduced chitosan-intercalated nontronite. *Environmental Science & Technology* **2020,** *54*, (8), 5207–5217.

35. Guo, D.; Xia, Q.; Zeng, Q.; Wang, X.; Dong, H., Antibacterial Mechanisms of Reduced Iron-Containing Smectite–Illite Clay Minerals. *Environment Science & Technology* **2021,** *55*, (22), 15256-15265.

36. Chen, R.; Liu, H.; Tong, M.; Zhao, L.; Zhang, P.; Liu, D.; Yuan, S., Impact of Fe (II) oxidation in the presence of iron-reducing bacteria on subsequent Fe (III) bio-reduction. *Science of the Total Environment* **2018,** *639*, 1007-1014.

37. Nivière, V.; Fontecave, M., Discovery of superoxide reductase: an historical perspective. *JBIC Journal of Biological Inorganic Chemistry* **2004,** *9*, (2), 119-123.

38. Merino, C.; Matus, F.; Kuzyakov, Y.; Dyckmans, J.; Stock, S.; Dippold, M. A., Contribution of the Fenton reaction and ligninolytic enzymes to soil organic matter mineralisation under anoxic conditions. *Science of the Total Environment* **2021,** *760*.

39. Hall, S. J.; Silver, W. L., Iron oxidation stimulates organic matter decomposition in humid tropical forest soils. *Global Change Biology* **2013,** *19*, (9), 2804-2813.

40. Van Bodegom, P. M.; Broekman, R.; Van Dijk, J.; Bakker, C.; Aerts, R., Ferrous iron stimulates phenol oxidase activity and organic matter decomposition in waterlogged wetlands. *Biogeochemistry* **2005,** *76*, (1), 69-83.

41. Egli, C. M.; Janssen, E., Proteomics approach to trace site-specific damage in aquatic extracellular enzymes during photoinactivation. *Environment Science & Technology* **2018,** *52*, (14), 7671-7679.

42. Egli, C. M.; Stravs, M. A.; Janssen, E. M. L., Inactivation and site-specific oxidation of aquatic extracellular bacterial leucine aminopeptidase by singlet oxygen. *Environment Science & Technology*

2020, *54*, (22), 14403-14412.

43. Moscatelli, M.; Lagomarsino, A.; Garzillo, A.; Pignataro, A.; Grego, S., β-Glucosidase kinetic parameters as indicators of soil quality under conventional and organic cropping systems applying two analytical approaches. *Ecological Indicators* **2012,** *13*, (1), 322-327.

44. Chacon, S. S.; Garcia-Jaramillo, M.; Liu, S. Y.; Ahmed, M.; Kleber, M., Differential capacity of kaolinite and birnessite to protect surface associated proteins against thermal degradation. *Soil Biology* and *Biochemistry* **2018,** *119*, 101-109.

45. Geiger, G.; Brandl, H.; Furrer, G.; Schulin, R. J. S. B.; Biochemistry, The effect of copper on the activity of cellulase and β-glucosidase in the presence of montmorillonite or Al-montmorillonite. *Soil Biology and Biochemistry* **1998,** *30*, (12), 1537-1544.

46. Turner, B. L.; Hopkins, D. W.; Haygarth, P. M.; Ostle, N., beta-glucosidase activity in pasture soils. *Appl. Soil Ecol.* **2002,** *20*, (2), 157-162.

47. Haneda, K.; Morrish, A., Magnetite to maghemite transformation in ultrafine particles. *Le Journal de Physique Colloques* **1977,** *38*, (C1), C1-321-C1-323.

48. Hayano, K.; Tubaki, K., Origin and properties of β-glucosidase activity of tomato-field soil. *Soil Biology Biochemistry* **1985,** *17*, (4), 553-557.

49. Pestovsky, O.; Stoian, S.; Bominaar, E. L.; Shan, X.; Münck, E.; Que Jr, L.; Bakac, A., Aqueous FeIV=O: Spectroscopic Identification and Oxo-Group Exchange. *Angewandte Chemie* **2005,** *117*, (42), 7031-7034.

50. Adler, P.; Eriksson, S., Structural properties, Mössbauer spectra, and magnetism of perovskite-type oxides SrFe1–xTixO3–y. *Zeitschrift für anorganische und allgemeine Chemie* **2000,** *626*, (1), 118-124.

51. Murad, E.; Cashion, J., *Mössbauer spectroscopy of environmental materials and their industrial utilization*. Springer Science & Business Media: 2011.

52. Weintraub, M. N.; Schimel, J. P., Seasonal protein dynamics in Alaskan arctic tundra soils. *Soil Biology and Biochemstry* **2005,** *37*, (8), 1469-1475.

53. Tietjen, T.; Wetzel, R. G., Extracellular enzyme-clay mineral complexes: Enzyme adsorption, alteration of enzyme activity, and protection from photodegradation. *Aquatic Ecology* **2003,** *37*, (4), 331-339.

54. Yang, Z. M.; Liao, Y. J.; Fu, X.; Zaporski, J.; Peters, S.; Jamison, M.; Liu, Y. R.; Wullschleger, S. D.; Graham, D. E.; Gu, B. H., Temperature sensitivity of mineral-enzyme interactions on the hydrolysis of cellobiose and indican by beta-glucosidase. *Science of the Total Environment* **2019,** *686*, 1194-1201.

55. Zang, Y.; Liu, F.; Li, X. X.; Sheng, A. X.; Zhai, J. Y.; Liu, J., Adsorption kinetics, conformational change, and enzymatic activity of beta-glucosidase on hematite (alpha-Fe_2O_3) surfaces. *Colloids and Surfaces B-Biointerfaces* **2020,** *193*.

56. Dick, R. P.; Dick, L. K.; Deng, S. P.; Li, X. F.; Kandeler, E.; Poll, C.; Freeman, C.; Jones, T. G.; Weintraub, M. N.; Esseili, K. A.; Saxena, J., Cross-laboratory comparison of fluorimetric microplate and colorimetric bench-scale soil enzyme assays. *Soil Biology and Biochemistry* **2018,** *121*, 240-248.

57. McGinnis, M. L.; Holub, S. M.; Myrold, D. D., Regional assessment of soil microbial functional diversity of Douglas-fir forests. *Soil Science Society of America Journal* **2014,** *78*, (S1), S125-S135.

58. Zimmerman, A. R.; Ahn, M.-Y., Organo-mineral–enzyme interaction and soil enzyme activity. In *Soil Enzymology*, Springer: 2010; pp 271-292.

59. Lammirato, C.; Miltner, A.; Wick, L. Y.; Kastner, M., Hydrolysis of cellobiose by beta-glucosidase in the presence of soil minerals - Interactions at solid-liquid interfaces and effects on enzyme activity

levels. *Soil Biology and Biochemstry* **2010,** *42*, (12), 2203-2210.

60. Gopinath, S.; Sugunan, S., Enzymes immobilized on montmorillonite K 10: effect of adsorption and grafting on the surface properties and the enzyme activity. *Applied Clay Science* **2007,** *35*, (1-2), 67-75.

61. Fusi, P.; Ristori, G.; Calamai, L.; Stotzky, G., Adsorption and binding of protein on “clean” (homoionic) and “dirty” (coated with Fe oxyhydroxides) montmorillonite, illite and kaolinite. *Soil Biology Biochemistry* **1989,** *21*, (7), 911-920.

62. Grover, A. K.; Macmurchie, D. D.; Cushley, R. J., Studies on almond emulsin Beta-D-glucosidase. 1. Isolation and characterization of a bifunctional isoenzyme. *Biochimica et Biophysica Acta* **1977,** *482*, (1), 98-108.

63. Gómez, J.; Romero, M.; Fernández, T.; García, S., Immobilization and enzymatic activity of β-glucosidase on mesoporous SBA-15 silica. *Journal of Porous Materials* **2010,** *17*, (6), 657-662.

64. Liu, F.; Li, X. X.; Sheng, A. X.; Shang, J. Y.; Wang, Z. M.; Liu, J., Kinetics and mechanisms of protein adsorption and conformational change on hematite particles. *Environment Science & Technology* **2019,** *53*, (17), 10157-10165.

65. Gournis, D.; Karakassides, M.; Petridis, D., Formation of hydroxyl radicals catalyzed by clay surfaces. *Physics and Chemistry of Minerals* **2002,** *29*, (2), 155-158.

66. Schimel, J.; Becerra, C. A.; Blankinship, J., Estimating decay dynamics for enzyme activities in soils from different ecosystems. *Soil Biology and Biochemistry* **2017,** *114*, 5-11.

67. Chen, Y.; Miller, C. J.; Waite, T. D., Heterogeneous Fenton chemistry revisited: Mechanistic insights from ferrihydrite-mediated oxidation of formate and oxalate. *Environment Science & Technology* **2021,** *55*, (21), 14414-14425.

68. Wardman, P., Reduction potentials of one-electron couples involving free radicals in aqueous solution. *Journal of Physical and Chemical Reference Data* **1989,** *18*, (4), 1637-1755.

69. Hall, S. J.; Treffkorn, J.; Silver, W. L., Breaking the enzymatic latch: impacts of reducing conditions on hydrolytic enzyme activity in tropical forest soils. *Ecology* **2014,** *95*, (10), 2964-2973.

70. Riedel, T.; Zak, D.; Biester, H.; Dittmar, T., Iron traps terrestrially derived dissolved organic matter at redox interfaces. *Proceedings of the National Academy of Sciences* **2013,** *110*, (25), 10101-10105.

71. Freeman, C.; Ostle, N.; Kang, H., An enzymic 'latch' on a global carbon store - A shortage of oxygen locks up carbon in peatlands by restraining a single enzyme. *Nature* **2001,** *409*, (6817), 149-149.

72. Hall, S. J.; Ye, C.; Weintraub, S. R.; Hockaday, W. C., Molecular trade-offs in soil organic carbon composition at continental scale. *Nature Geoscience* **2020,** *13*, (10), 687-692.

73. Gallagher, W., FTIR Analysis of Protein Structure. *Soil Sci. Soc. Am. J.* **2009,** *455*.

74. Trusiak, A.; Treibergs, L. A.; Kling, G. W.; Cory, R. M., The role of iron and reactive oxygen species in the production of CO2 in arctic soil waters. *Geochimica et Cosmochimica Acta* **2018,** *224*, 80-95.

75. Murad, E.; Fischer, W. R., *Iron in Soils and Clay Minerals*. Springer Netherlands: Dordrecht: 1988.

76. Favre, F.; Stucki, J. W.; Boivin, P., Redox properties of structural Fe in ferruginous smectite. A discussion of the standard potential and its environmental implications. *Clays and Clay Minerals* **2006,** *54*, (4), 466-472.

77. Canfield, D. E.; Thamdrup, B.; Hansen, J. W. J. G. e. C. A., The anaerobic degradation of organic matter in Danish coastal sediments: iron reduction, manganese reduction, and sulfate reduction. *Geochimica et Cosmochimica Acta* **1993,** *57*, (16), 3867-3883.

78. Lützow, M. v.; Kögel-Knabner, I.; Ekschmitt, K.; Matzner, E.; Guggenberger, G.; Marschner, B.;

Flessa, H., Stabilization of organic matter in temperate soils: mechanisms and their relevance under different soil conditions–a review. *European Journal of Soil Science* **2006,** *57*, (4), 426-445.

FIGURE CAPTIONS

Figure 1 Time-course change of total Fe(II) at pH 5 (a) and pH 7 (b) under both anoxic and oxic conditions. Time-course production of •OH at pH 5 (c) and pH 7 (d) under both anoxic and oxic conditions. Significant difference was observed between anoxic and oxic conditions (t-test, p<0.001, Table S3). Positive correlations between the concentrations of •OH and the total Fe(II) oxidation extent (ΔTotal Fe) at pH 5 (e) and pH 7 (f). Significant linear correlations were shown (p< 0.05). ΔTotal Fe(II)=$Fe(II)_{initial}$–$Fe(II)_t$, where $Fe(II)_{initial}$ represents the concentration of Fe(II) at time 0, and $Fe(II)_t$ refers to the content of Fe(II) at time t. Error bars represent one standard deviation from triplicate experiments.

Figure 2 Mass-normalized total BG activity ($Activity_{total}$) including both supernatant and mineral-adsorbed BG in nontronite rNAu-2 at pH 5 after 2-hour (a) and 24-hour (b) or at pH 7 after 2-hour (c) and 24-hour (d) under both anoxic and oxic conditions. The free enzyme activity was shown under the same conditions. Error bars represent one standard deviation from triplicate experiments. Difference analysis in BG activity between different experimental groups was shown in Table S4.

Figure 3 Pearson linear correlations of the extent of inhibition of total BG activity ($Activity_{oxic}$/$Activity_{anoxic}$) versus the Fe(II) oxidation extent (a) and the cumulative amount of •OH (b), including the data obtained from 2 h and 24 h incubations at both pH 5 and 7 conditions (significant correlations were shown, p<0.05). The changes of

BG activity after 2 h and 24 h in the presence of rNAu-2 at both pH 5 (c) and pH 7 (d) under both anoxic and oxic conditions, and under oxic condition augmented with SB (•OH scavenger), catalase (H_2O_2 scavenger), TEMPO (•O^{2-} scavenger). BG concentration is 2 mg/L. The shadings in a-b represent 95% confidence intervals. Error bars represent one standard deviation from triplicate experiments. Significant differences in BG activity between selected experiment groups were shown (t-test, * p<0.05, * p<0.01, *** p<0.001) (Table S5).

Figure 4. (a) ATR-FTIR spectra of free BG including the characteristic bands of Amide I, Amide II, Amide III, and side-chain δ CH_2 and υs COO-. (b) Deconvolution of Fourier transform infrared spectroscopy Amide I band into four secondary structures. (c) Relative peak intensity of 4 secondary structures of the Amide I band for mineral-adsorbed BG (BG in the mineral-free experiment was shown as “control”), and in some experiments augmented with SB (•OH scavenger), catalase (H_2O_2 scavenger), and TEMPO (•O^{2-} scavenger) at pH 5. The data of BG adsorbed to unreduced minnerals were also shown. All samples were under oxic condition. Error bars represent one standard deviation from duplicate experiments.

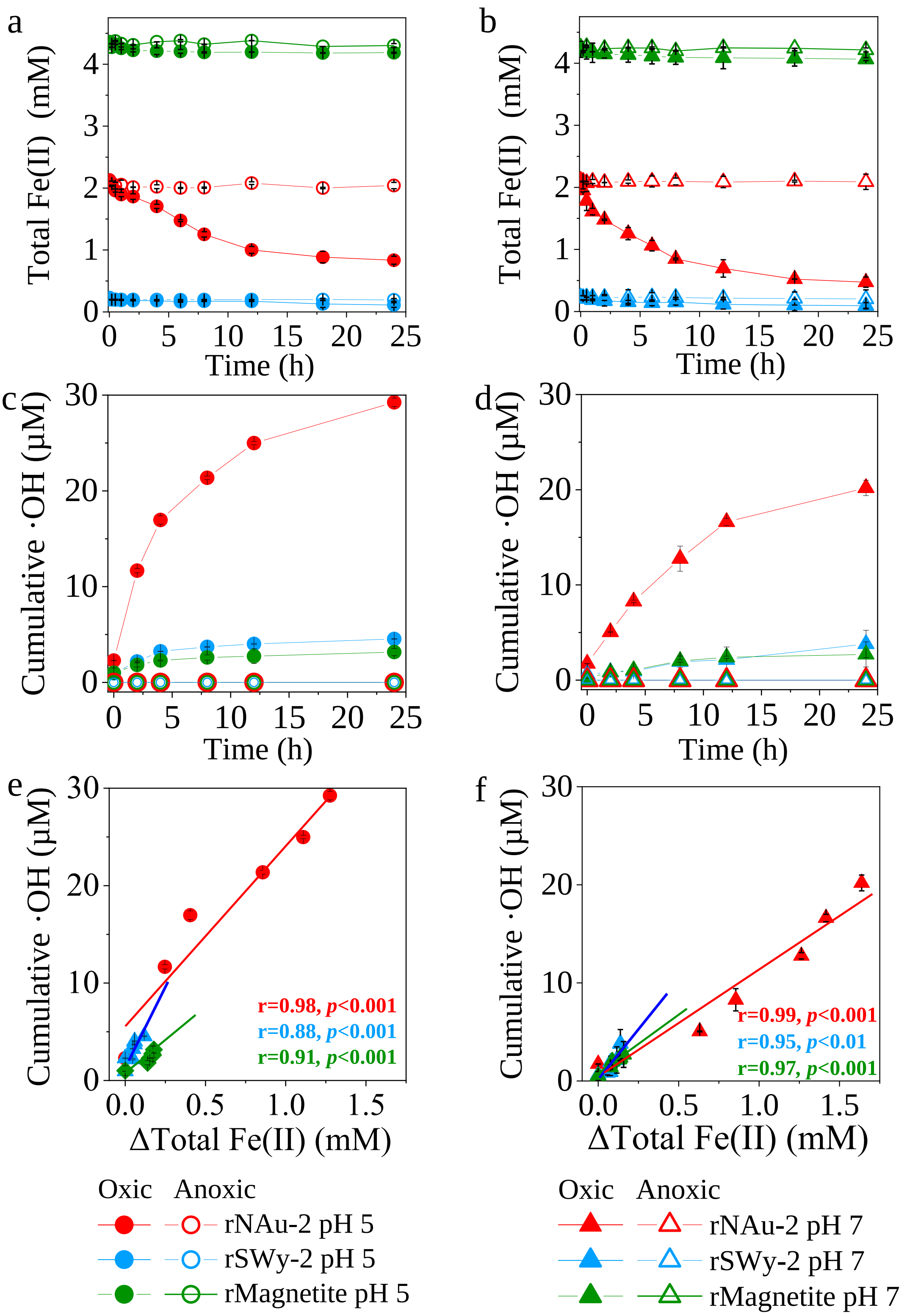
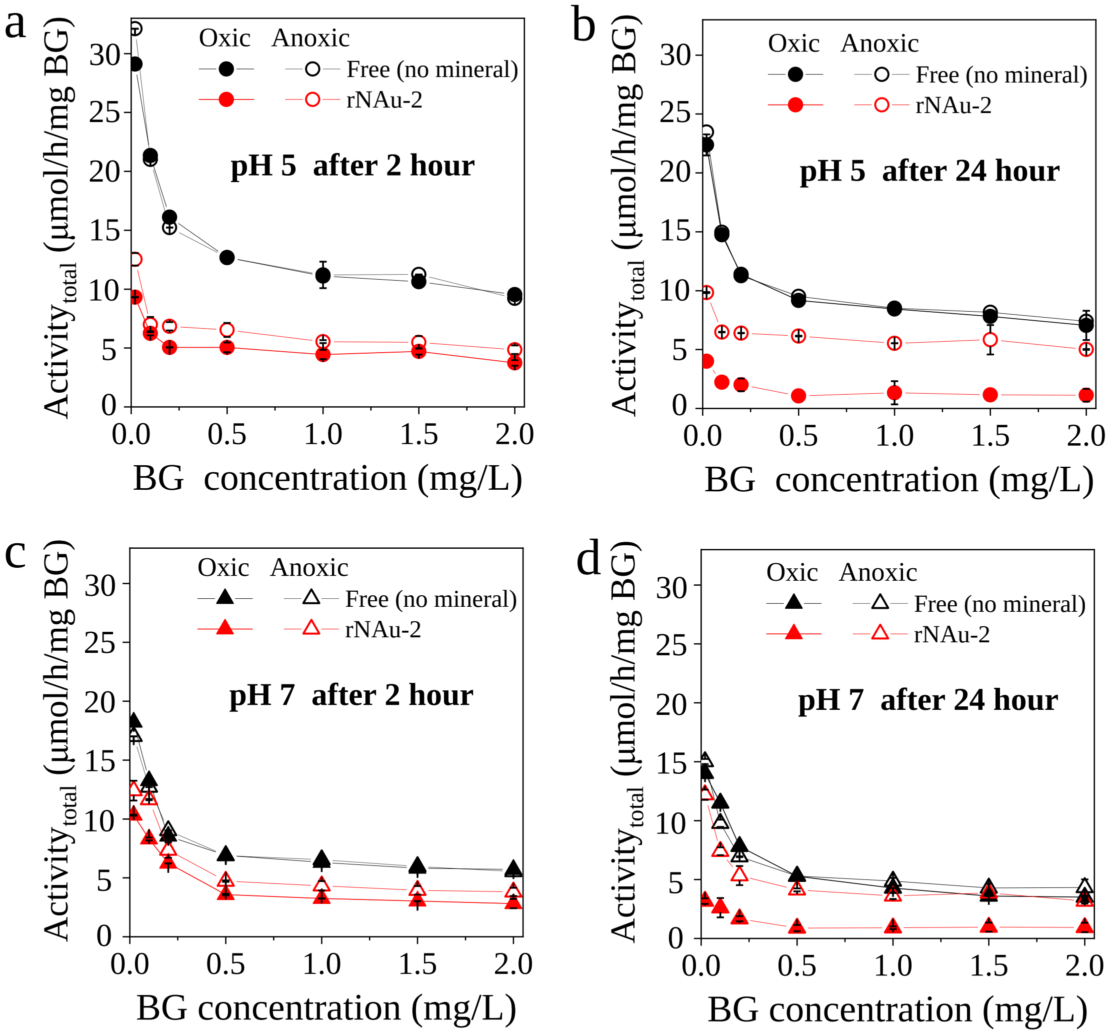

a
b
c
d
e
f
Total Fe(II) (mM)
Time (h)
Cumulative ·OH (μM)
ΔTotal Fe(II) (mM)
r=0.98, p<0.001
r=0.88, p<0.001
r=0.91, p<0.001
r=0.99, p<0.001
r=0.95, p<0.01
r=0.97, p<0.001
Oxic Anoxic
rNAu-2 pH 5
rSWy-2 pH 5
rMagnetite pH 5
rNAu-2 pH 7
rSWy-2 pH 7
rMagnetite pH 7

Figure 1



a
b
c
d
Oxic Anoxic
Free (no mineral)
rNAu-2
pH 5 after 2 hour
pH 5 after 24 hour
pH 7 after 2 hour
pH 7 after 24 hour
Activity$_{total}$ (μmol/h/mg BG)
BG concentration (mg/L)
0
5
10
15
20
25
30
0.0
0.5
1.0
1.5
2.0

Figure 2

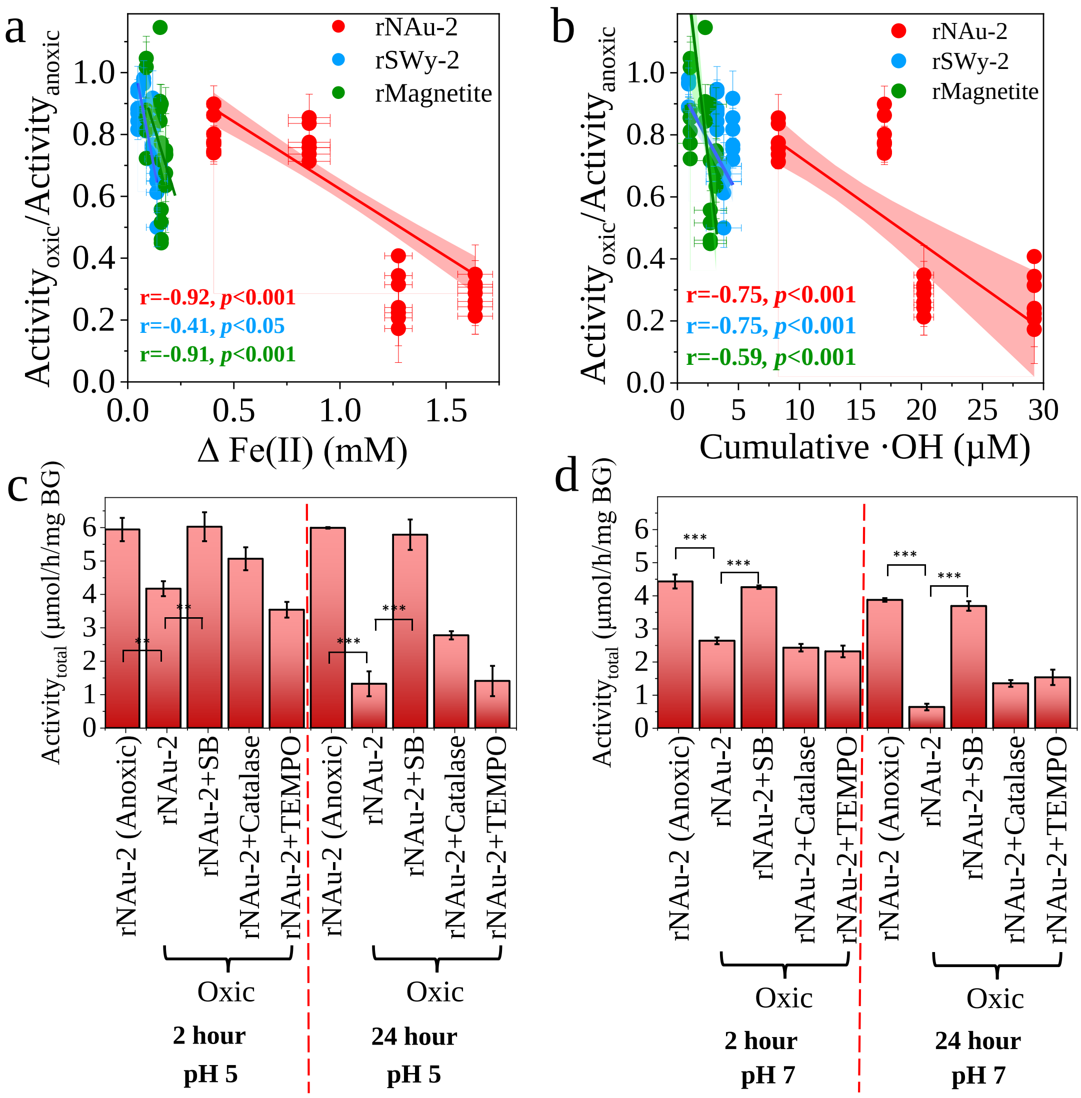
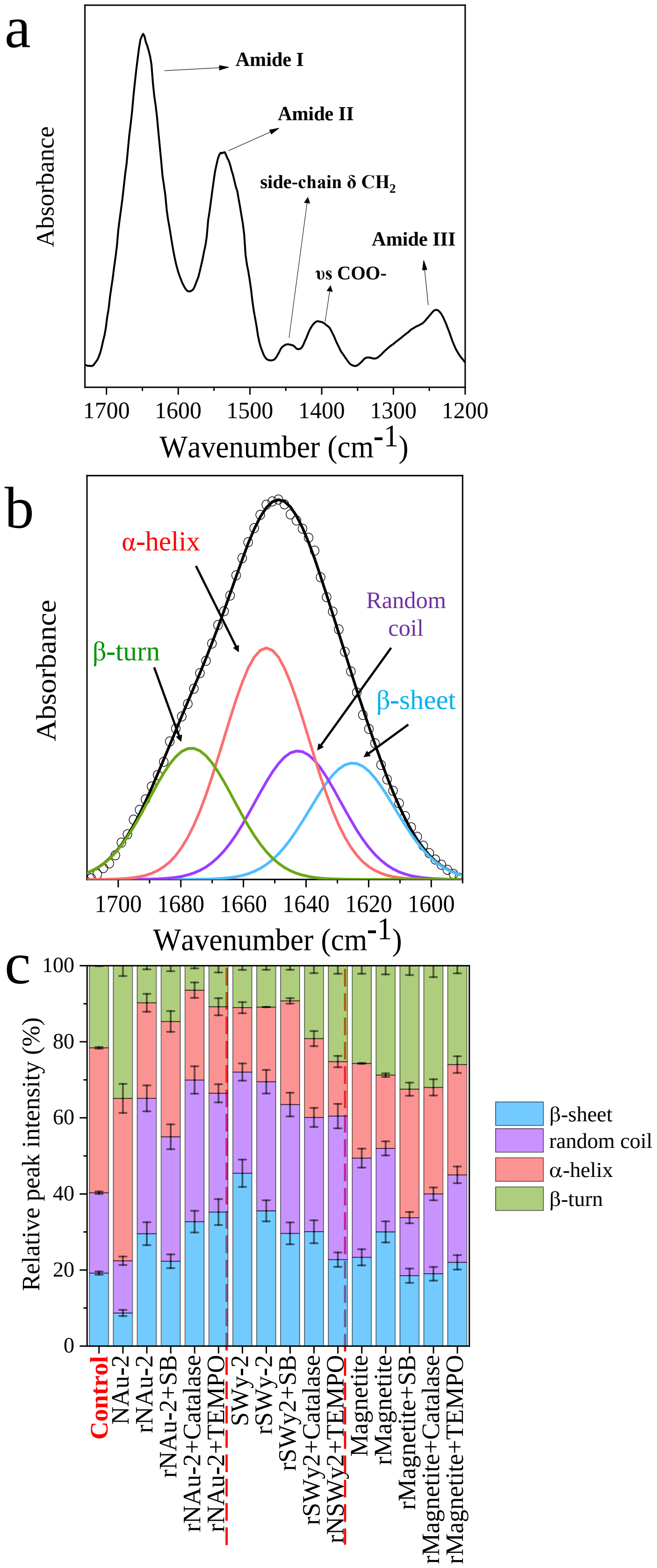

a
rNAu-2
rSWy-2
rMagnetite
Activity$_{oxic}$/Activity$_{anoxic}$
r=-0.92, p<0.001
r=-0.41, p<0.05
r=-0.91, p<0.001
Δ Fe(II) (mM)
b
rNAu-2
rSWy-2
rMagnetite
Activity$_{oxic}$/Activity$_{anoxic}$
r=-0.75, p<0.001
r=-0.75, p<0.001
r=-0.59, p<0.001
Cumulative ·OH (μM)
c
Activity$_{total}$ (μmol/h/mg BG)
rNAu-2 (Anoxic)
rNAu-2
rNAu-2+SB
rNAu-2+Catalase
rNAu-2+TEMPO
Oxic
2 hour
pH 5
24 hour
pH 5
d
Activity$_{total}$ (μmol/h/mg BG)
Oxic
2 hour
pH 7
24 hour
pH 7

Figure 3



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698 **Figure 4**